

CONSTANT SPEED STORY: 3-STAGE DUAL-GRAPH EDITION

UNIT 2: KINEMATICS • 3-STAGE JOURNEY MODELING • DAY 15 PERFORMANCE TASK

Name: _____ Period: _____ Date: _____

SCORE: _____ / 10 pts

Mission: Invent an original character/vehicle journey across **3 distinct motion sections** over **10.0 seconds** (like the Dual-Graph Studio). Illustrate the scene, author your narrative, and calculate the velocities in the table. On the back, prove your math with both graphs!

HS-PS2-1 • PERFORMANCE TASK

PART 1: 3-STAGE JOURNEY ILLUSTRATION

Sketch the object, path of motion, origin ($x = 0\text{m}$), pause/turning point, and final point

[Draw your character, vehicle, drone, rover, or athlete traveling across 3 distinct stages]
Label: Starting Reference Point (x_0) • Section 1 Path ($\rightarrow$ or $\leftarrow$) • Section 2 Stop/Action Point • Section 3 Final Destination

PART 2: ORIGINAL 3-SECTION MOTION NARRATIVE

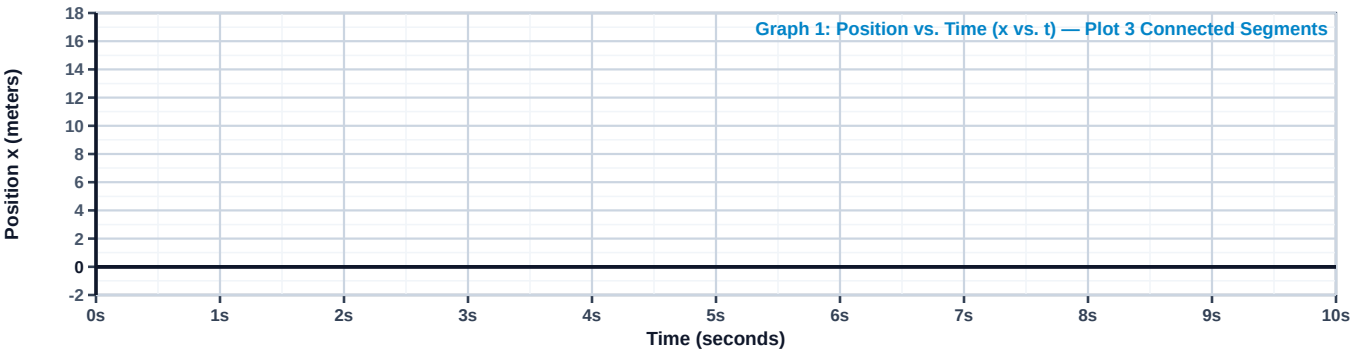
Describe what occurs in each phase (Total Time = 10 seconds)

Section 1 (0s to t_1) Initial Motion Starts at $x =$ _____m, travels to $x =$ _____m: _____ _____ _____	Section 2 (t_1 to t_2) Mid-Course Action Action (e.g. stopped/waiting/steady): _____ _____ _____	Section 3 (t_2 to 10s) Final Stretch Travels from $x =$ _____m to $x =$ _____m: _____ _____ _____
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PART 3: 3-STAGE MOTION BREAKDOWN & G.U.E.S.S. MATHEMATICAL PROOF

Compute displacement (Δx) and velocity ($v = \Delta x / \Delta t$) for each section

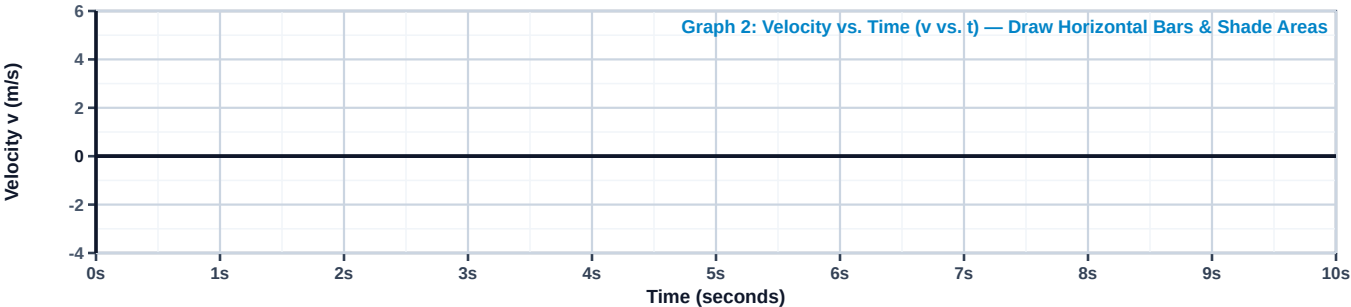
Measurement / Step	Section 1	Section 2	Section 3	Whole Trip (Total)
Time Interval (Δt) $\Delta t = t_{\text{final}} - t_{\text{initial}}$	From ____s to ____s $\Delta t_1 =$ ____ s	From ____s to ____s $\Delta t_2 =$ ____ s	From ____s to 10s $\Delta t_3 =$ ____ s	$\Delta t_{\text{total}} = 10.0 \text{ s}$
Position Coordinates Initial (x_i) $\rightarrow$ Final (x_f)	$x_i =$ ____ m $x_f =$ ____ m	$x_i =$ ____ m $x_f =$ ____ m	$x_i =$ ____ m $x_f =$ ____ m	Start $x_0 =$ ____ m Final $x_{\text{end}} =$ ____ m
Displacement (Δx) $\Delta x = x_f - x_i$ (include + / -)	$\Delta x_1 =$ ____ m	$\Delta x_2 =$ ____ m	$\Delta x_3 =$ ____ m	Net $\Delta x =$ ____ m
Direction of Motion	☐ Forward (+) ☐ Stopped (0) ☐ Backward (-)	☐ Forward (+) ☐ Stopped (0) ☐ Backward (-)	☐ Forward (+) ☐ Stopped (0) ☐ Backward (-)	Total Distance: $d_{\text{total}} =$ ____ m
G.U.E.S.S. Velocity Formula $v = \Delta x / \Delta t$ (Substitute & Solve)	$v_1 =$ ____m / ____s $v_1 =$ ____ m/s	$v_2 =$ ____m / ____s $v_2 =$ ____ m/s	$v_3 =$ ____m / ____s $v_3 =$ ____ m/s	Average Speed: $v_{\text{avg}} = d_{\text{tot}} / 10\text{s}$ $=$ ____ m/s



Section 1 Slope Proof:
 $m_1 = \Delta x_1 / \Delta t_1 = (\text{____} \text{m}) / (\text{____} \text{s})$
Slope₁ = ____ m/s (Matches v_1 ?)

Section 2 Slope Proof:
 $m_2 = \Delta x_2 / \Delta t_2 = (\text{____} \text{m}) / (\text{____} \text{s})$
Slope₂ = ____ m/s (Flat line = 0?)

Section 3 Slope Proof:
 $m_3 = \Delta x_3 / \Delta t_3 = (\text{____} \text{m}) / (\text{____} \text{s})$
Slope₃ = ____ m/s (Matches v_3 ?)



Section 1 Area (Δx_1):
 $\text{Area}_1 = \text{Height} \times \text{Width} = (v_1)(\Delta t_1)$
= (____ m/s) $\times$ (____ s) = ____ m

Section 2 Area (Δx_2):
 $\text{Area}_2 = \text{Height} \times \text{Width} = (v_2)(\Delta t_2)$
= (0 m/s) $\times$ (____ s) = 0 m

Section 3 Area (Δx_3):
 $\text{Area}_3 = \text{Height} \times \text{Width} = (v_3)(\Delta t_3)$
= (____ m/s) $\times$ (____ s) = ____ m

PART 4: DUAL-GRAPH SYNTHESIS & NET DISPLACEMENT VERIFICATION

CORE CONCEPT CHECK

1. Total Net Displacement Proof from Shaded Areas:

Sum of all 3 areas: $\Delta x_{\text{total}} = \text{Area}_1 + \text{Area}_2 + \text{Area}_3$

= (____ m) + (0 m) + (____ m) = ____ meters

Does this match your position change on Graph 1 ($x_{\text{final}} - x_{\text{initial}}$)? ☐ YES ☐ NO

2. Physical Meaning of Negative Slope / Area:

When a line on Graph 1 slopes downward (negative slope), what direction is the object moving? _____

Where does its horizontal line appear on Graph 2? _____

3. Unit Cancellation Proof: In Graph 2, show mathematically how multiplying the height unit (m/s) by the width unit (s) cancels to prove that area MUST be distance: (m / s) $\times$ (s) = _____